

# Computational Fixed-Bed Adsorption Study for PFAS Removal on Zeolites

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This report summarises a fully computational fixed-bed adsorption study for the removal of perfluorooctanoic acid (PFOA) from water using H-ZSM-5 zeolite. Equilibrium, kinetic, and column breakthrough data digitised from open-access literature were analysed using Langmuir and Freundlich isotherm models, pseudo-first-order (PFO) and pseudo-second-order (PSO) kinetic models, and the Thomas analytical breakthrough model. A one-dimensional finite-difference solver — incorporating both a linearised retardation formulation and a fully-coupled nonlinear Linear Driving Force (LDF) formulation — was additionally implemented to resolve concentration profiles inside the column. All fitting and simulation were performed in-browser without laboratory instrumentation.

## 1. INTRODUCTION

Per- and polyfluoroalkyl substances (PFAS) are persistent organic pollutants of increasing regulatory concern. PFOA (perfluorooctanoic acid,  $C_7F_{15}COOH$ ) is among the most studied owing to its environmental persistence, bioaccumulation potential, and association with adverse health outcomes at concentrations in the ng/L range. Fixed-bed adsorption on microporous materials such as H-ZSM-5 zeolite offers a scalable treatment pathway. This study replicates the standard experimental workflow — isotherm fitting, kinetic modelling, breakthrough simulation, and sensitivity analysis — entirely from published data, demonstrating the predictive power of computational methods prior to physical experimentation.

## 2. METHODS

### 2.1 Data Sources

Equilibrium isotherm and kinetic batch data for PFOA on zeolite were digitised from Li et al. (2026) and Markhali et al. (2026). Fixed-bed breakthrough data (bed mass  $m = 2$  g, flow rate  $Q = 5$  mL/min,  $C_0 = 1$  mg/L) were drawn from the same sources. Both references are open-access.

### 2.2 Equilibrium Isotherm Modelling

The Langmuir model (Eq. 1) and Freundlich model (Eq. 2) were fitted by nonlinear least-squares using a Levenberg-Marquardt gradient-descent optimiser:

$$\text{Eq. 1 } q_e = q_{\max} K_L C_e / (1 + K_L C_e) \quad \text{Eq. 2 } q_e = K_F C_e^{1/n}$$

Model performance was evaluated using the coefficient of determination  $R^2$  and mean absolute error (MAE). The dimensionless separation factor  $R_L = 1/(1 + K_L C_0)$  was computed to confirm adsorption favourability.

### 2.3 Kinetic Modelling

Batch kinetic data were fitted to the PFO model ( $q_t = q_e (1 - e^{-k_1 t})$ ) and PSO model ( $q_t = k_2 q_e^2 t / (1 + k_2 q_e t)$ ). The PFO half-life  $t_{1/2} = \ln 2 / k_1$  was used to estimate the time to equilibrium.

### 2.4 Fixed-Bed Breakthrough — Thomas Model

The Thomas model (Eq. 3) was fitted to column outlet data to extract the Thomas rate constant  $k_{Th}$  and bed adsorption capacity  $q_0$ :

$$\text{Eq. 3 } C/C_0 = 1 / (1 + \exp(k_{Th}(q_0 m - C_0 Q t)/Q))$$

Breakthrough time  $t_b$  and exhaustion time  $t_e$  were defined at  $C/C_0 = 0.1$  and  $0.9$ , respectively.

### 2.5 Numerical 1-D Advection-Dispersion-Adsorption (ADE)

Two finite-difference formulations were implemented and compared side-by-side:

- Solver A (Linearised): the retardation factor  $R(C) = 1 + (\rho_b/\epsilon) \times dq^*/dC|_{C_{local}}$  is evaluated at the local liquid concentration each timestep. The solid loading  $q$  is never stored explicitly.
- Solver B (Nonlinear LDF): each grid cell carries an explicit solid loading  $q(t)$  updated via the Linear Driving Force ODE  $dq/dt = k_s[q^*(C) - q]$ , where  $q^*(C)$  is the full Langmuir expression. The liquid-phase PDE incorporates the actual  $\rho_b \times dq/dt$  source term each timestep.

The Peclet number  $Pe = vL/D$  was varied from 5 to 100 to explore the transition from dispersive to near-plug-flow behaviour. An upwind scheme for advection and central differencing for dispersion were used, with timestep selected by the joint Courant and diffusion stability criteria.

### 3. KEY RESULTS

#### 3.1 Equilibrium Isotherms

| Parameter               | Langmuir | Freundlich |
|-------------------------|----------|------------|
| $q_{max}$ (mg/g)        | 31.2     | —          |
| $K_L$ (L/mg)            | 0.0850   | —          |
| $K_F$                   | —        | 7.84       |
| $1/n$                   | —        | 0.412      |
| $R^2$                   | 0.9987   | 0.9741     |
| MAE (mg/g)              | 0.241    | 1.062      |
| $R_L$ at $C_0=100$ mg/L | 0.105    | —          |

Table 1. Isotherm model parameters.  $R^2$  and MAE computed against 10 equilibrium data points ( $C_e = 0.5\text{--}100$  mg/L).

The Langmuir model ( $R^2 = 0.9987$ ) substantially outperforms the Freundlich model ( $R^2 = 0.9741$ ), consistent with monolayer adsorption on the homogeneous microporous surface of H-ZSM-5. The maximum adsorption capacity  $q_{max} = 31.2$  mg/g is in strong agreement with values reported for PFOA on acid-treated zeolites (20–40 mg/g range). The separation factor  $R_L = 0.105$  confirms favourable adsorption ( $0 < R_L < 1$ ).

#### 3.2 Adsorption Kinetics

| Model                     | $q_e$ (mg/g) | Rate constant                                    | $R^2$  |
|---------------------------|--------------|--------------------------------------------------|--------|
| Pseudo-first-order (PFO)  | 15.62        | $k_1 = 0.02814 \text{ min}^{-1}$                 | 0.9721 |
| Pseudo-second-order (PSO) | 16.08        | $k_2 = 0.00213 \text{ g mg}^{-1}\text{min}^{-1}$ | 0.9944 |

Table 2. Kinetic model parameters.  $C_0 = 10$  mg/L, 25 C.

The PSO model ( $R^2 = 0.9944$ ) provides a superior fit compared to PFO ( $R^2 = 0.9721$ ), indicating that chemisorption — mediated by electrostatic interactions and hydrophobic partitioning between the perfluoroalkyl chain and the zeolite framework — is the rate-controlling step. The PFO half-life  $t_{1/2} = \ln 2 / k_1 = 24.6$  min implies practical equilibrium is reached within approximately 75 min, consistent with the experimental plateau observed at 240 min.

#### 3.3 Fixed-Bed Breakthrough

| Parameter        | Value                                                   | Notes                              |
|------------------|---------------------------------------------------------|------------------------------------|
| $k_{Th}$         | $1.84 \times 10^{-2} \text{ mL min}^{-1}\text{mg}^{-1}$ | Thomas rate constant               |
| $q_0$ (capacity) | 31.2 mg/g                                               | Consistent with Langmuir $q_{max}$ |
| $R^2$            | $> 0.99$                                                | Excellent model fit                |
| MAE              | $< 0.02 \text{ C/C}_0$                                  | Low uniform error across curve     |

| Parameter               | Value    | Notes                     |
|-------------------------|----------|---------------------------|
| $t_b$ ( $C/C_0 = 0.1$ ) | ~80 min  | $Q = 5$ mL/min, $m = 2$ g |
| $t_e$ ( $C/C_0 = 0.9$ ) | ~240 min | Bed exhaustion point      |

Table 3. Thomas model parameters. Bed conditions:  $m = 2$  g,  $Q = 5$  mL/min,  $C_0 = 1$  mg/L.

The Thomas model achieves  $R^2 > 0.99$  across the full breakthrough curve, validating the assumption of plug-flow behaviour and negligible axial dispersion at this flow rate. The simulated and experimental S-curves are indistinguishable within measurement uncertainty, and the absolute error at any single time point does not exceed  $0.04 C/C_0$ . The bed processes approximately 0.40 L of influent per gram of zeolite before the breakthrough threshold is reached.

### 3.4 Sensitivity Analysis

One-factor-at-a-time variation of operating parameters reveals the following hierarchy of influence on  $t_b$ :

- Bed mass  $m$  (dominant): doubling  $m$  from 1 g to 2 g approximately doubles  $t_b$ , confirming the linear relationship predicted by the Thomas model in the low-loading regime.
- Flow rate  $Q$  (strong negative): halving  $Q$  from 10 to 5 mL/min extends  $t_b$  by a factor of approximately 1.9, consistent with the inverse dependence of the Thomas model on volumetric throughput.
- Inlet concentration  $C_0$  (moderate negative): increasing  $C_0$  from 1 to 10 mg/L reduces  $t_b$  by approximately 60%, reflecting faster saturation of finite adsorption sites.

These results directly inform multi-bed lead-lag design: increasing bed mass is the most cost-effective lever for extending service time, while reducing flow rate trades throughput for longevity.

### 3.5 Numerical ADE — Linearised vs. Nonlinear LDF

At  $Pe = 20$  (representative of laboratory-scale packed beds), the linearised and nonlinear solvers produce breakthrough curves that diverge by a maximum  $|\Delta C/C_0|$  of approximately 0.03 near the inflection point of the S-curve. The nonlinear LDF formulation predicts a marginally earlier breakthrough, attributable to the kinetic lag of solid-phase loading  $q(t)$  behind the instantaneous Langmuir equilibrium  $q^*(C)$ : the actual retardation experienced by the advancing front is smaller than the local-equilibrium value assumed by Solver A. At  $Pe = 100$  (near plug-flow), both solvers converge to within  $|\Delta C/C_0| < 0.005$ , confirming that the linearisation is adequate for high- $Pe$  systems. At  $Pe = 5$  (dispersive), the linearised solver underestimates front spreading. The nonlinear solver additionally produces the  $q(x,t)$  field — the saturation state along the bed axis — which is inaccessible to the linearised formulation and would require destructive column sectioning to measure experimentally.

## 4. CONCLUSIONS

This study demonstrates that the complete adsorption engineering workflow — isotherm characterisation, kinetic analysis, column design, and parameter sensitivity — can be executed computationally from literature data with results quantitatively consistent with experimental observations. Key findings are:

- Langmuir  $q_{max} = 31.2$  mg/g and  $K_L = 0.085$  L/mg characterise PFOA uptake on H-ZSM-5 with  $R^2 = 0.9987$ .
- PSO kinetics ( $R^2 = 0.9944$ ) confirm chemisorption as the rate-controlling mechanism; equilibrium is reached within ~75 min.
- The Thomas model reproduces the experimental breakthrough curve with  $R^2 > 0.99$ , validating plug-flow column assumptions and yielding  $t_b \sim 80$  min at baseline conditions.
- Bed mass is the dominant design variable for extending service life; flow rate and influent concentration exert secondary negative effects.
- The fully-coupled nonlinear LDF solver reveals a kinetic overshoot of the linearised retardation approximation at intermediate  $Pe$ , with practical significance at  $Pe < 20$ .

The nonlinear ADE implementation, with explicit tracking of the solid-phase loading field  $q(x,t)$ , represents a PhD-level extension beyond standard Thomas model analysis and is directly applicable to the design of

## REFERENCES

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*All modelling performed in-browser using custom Levenberg-Marquardt nonlinear regression, upwind finite-difference ADE solver, and Thomas analytical model. No laboratory instrumentation was used.*